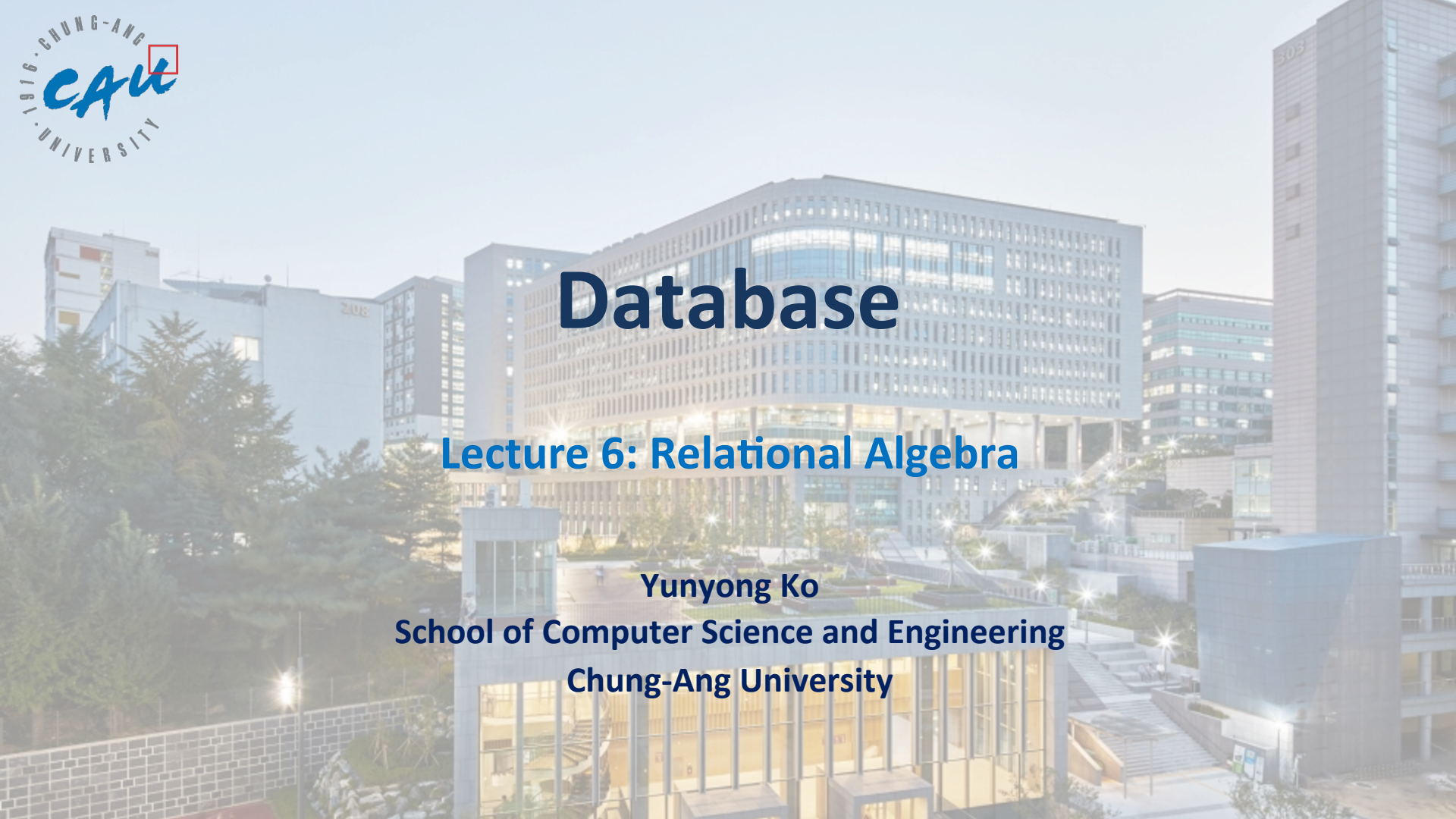


Database

Lecture 6: Relational Algebra

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Outline

- Introduction to Relational Algebra



- Unary Relational Operations
- Set Operations
- Binary Relational Operations
- Additional Relational Operations

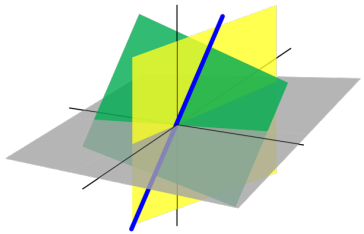
Introduction to Relational Algebra

What is the Algebra?

- Represent numbers and quantities in equations with letters and symbols
- Basic operations: +, -, *, /

What is the Linear Algebra?

- Linear transformation in vector spaces
- Basic operations
  - Vector addition and scalar multiplication, Matrix operations, …



What is the Relational Algebra?

- Theoretical and mathematical model for "*how to get data*" (i.e., procedural)
  - SQL is a high-level declarative language → *What data is needed?* (*don't care about how to get*)
- *The set of basic operations for the formal relational model*

Motivation of Relational Algebra

❑ Why Relational Algebra Important?

- Formal foundation for relational model operations
  - ❑ Selection, projection, renaming, join, cartesian product, …
  - ❑ Strictly based on the mathematical concepts of ***sets***

Relational Algebra Operators $\pi \sigma \rho$ $\bowtie \cup \cap$

```
SELECT   *
FROM     EMPLOYEE
WHERE    Dno=4 AND Salary>25000;
```

$$\sigma_{\text{Dno=4 AND Salary>25000}}(\text{EMPLOYEE})$$

```
SELECT   Pnumber, Dnum, Lname, Address, Bdate
FROM     ((PROJECT JOIN DEPARTMENT ON Dnum = Dnumber)
         JOIN EMPLOYEE ON Mgr_ssn = Ssn)
WHERE    Plocation = 'Stafford';
```

$$\text{STAFFORD_PROJS} \leftarrow \sigma_{\text{Plocation='Stafford'}}(\text{PROJECT})$$
$$\text{CONTR_DEPTS} \leftarrow (\text{STAFFORD_PROJS} \bowtie_{\text{Dnum=Dnumber}} \text{DEPARTMENT})$$
$$\text{PROJ_DEPT_MGRS} \leftarrow (\text{CONTR_DEPTS} \bowtie_{\text{Mgr_ssn=SsnE}} \text{MPLOYEE})$$
$$\text{RESULT} \leftarrow \pi_{\text{Pnumber, Dnum, Lname, Address, Bdate}}(\text{PROJ_DEPT_MGRS})$$

SQL: What data is needed?

Relational Algebra: How to get data?

- A basis for *query processing and optimization*
  - ❑ *i.e., specifying a "sequence" of operations to get the results*

Application of Relational Algebra

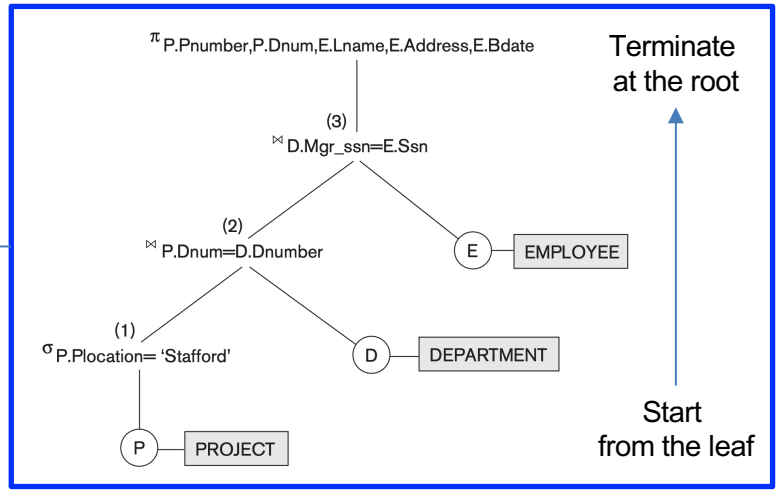
❑ What is a Query (Execution) Tree?

- Tree data structure for the *internal representation of the query* in DMBS
  - ❑ i.e., a relational algebra expression (a set of relational algebra operations to be executed)
  - ❑ **Leaf nodes**: the input relations, **Internal nodes**: the relational algebra operations

Q. For every project located in 'Stafford',
list the project number, the controlling department number,
and the department manager's last name, address, and birth date

```
SELECT   P.Pnumber, P.Dnum, E.Lname, E.Address, E.Bdate
FROM     PROJECT P, DEPARTMENT D, EMPLOYEE E
WHERE    P.Dnum=D.Dnumber AND D.Mgr_ssn=E.Ssn AND
         P.Plocation= 'Stafford';
```

STAFFORD_PROJS ← $\sigma_{Plocation='Stafford'}$(PROJECT)
CONTR_DEPTS ← (STAFFORD_PROJS $\bowtie_{Dnum=Dnumber}$ DEPARTMENT)
PROJ_DEPT_MGRS ← (CONTR_DEPTS $\bowtie_{Mgr_ssn=SsnE}$ MPLOYEE)
RESULT ← $\pi_{Pnumber, Dnum, Lname, Address, Bdate}$(PROJ_DEPT_MGRS)



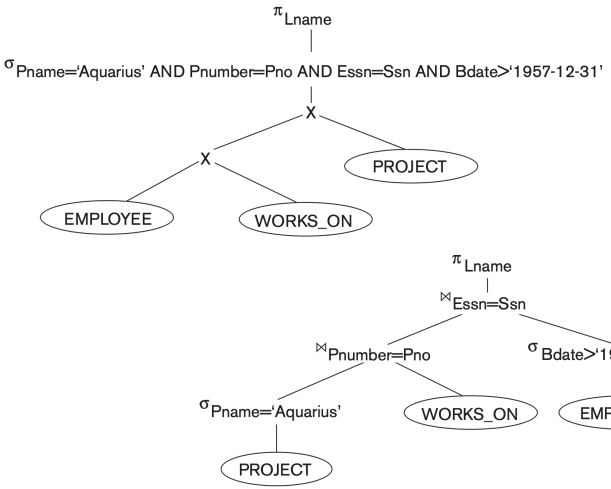
Explicitly representing the sequence of operations
→ A **good visual representation** and **understanding of the query**

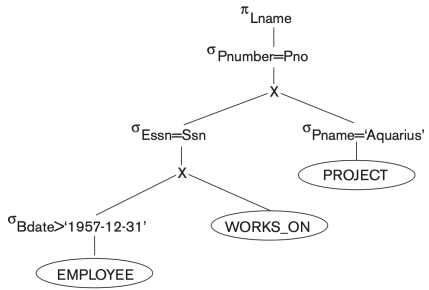
Application of Relational Algebra (Cont.)

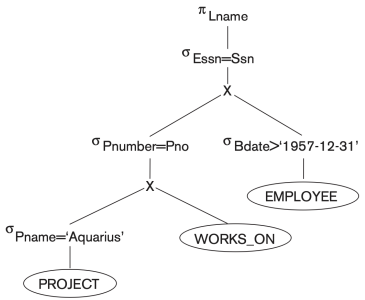
❑ What is Query Processing?

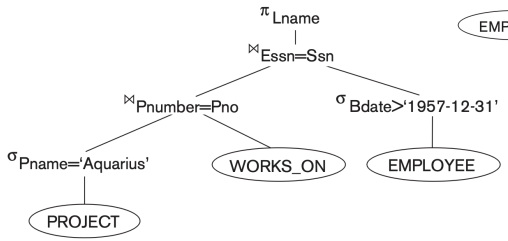
- The process of choosing a *suitable query tree of possible execution strategies*

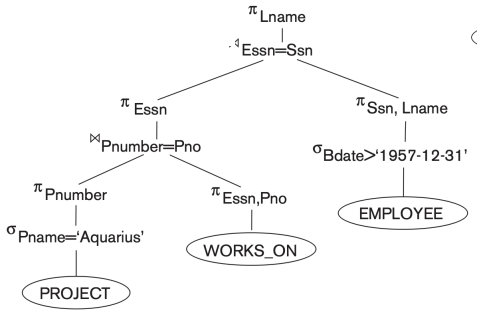
```
SELECT  E.Lname
FROM    EMPLOYEE E, WORKS_ON W, PROJECT P
WHERE   P.Pname='Aquarius' AND P.Pnumber=W.Pno AND E.Essn=W.Ssn
        AND E.Bdate > '1957-12-31';
```











Multiple query trees (plans) exist for a single SQL

Which one is the best?

→ Query Optimization in Week 13

Outline

- Introduction to Relational Algebra
- **Unary Relational Operations**
- **Set Operations**
- **Binary Relational Operations**
- **Additional Relational Operations**



Unary Relational Operations: SELECT

What is the SELECT Operation?

- To choose a *subset of the tuples* from "**a relation**" satisfying a selection condition
  - i.e., a **filter** that keeps only those tuples satisfying a qualifying condition, while discards the others

The SELECT operator (sigma) $\sigma_{\langle \text{selection condition} \rangle}(R)$ A relational algebra expression (relation)

A Boolean expression specified on the attributes of *R*

- Result of the SELECT Operation
  - The relation includes ***tuples satisfying the specified conditions*** and has the ***same attributes*** **as *R***
    - i.e., the degree of the resulting relation = the degree of *R*

$\sigma_{Dno=4}(EMPLOYEE)$

Applied to *each tuple individually*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Horizontal filtering

Selection Conditions of SELECT Operation

❑ The Boolean Expression of the SELECT Operation

- It is made up of a number of clauses of the form:

$$\sigma_{\text{<selection condition>}}(R)$$

<attribute name> <comparison op> <constant value>

or

<attribute name> <comparison op> <attribute name>

  - ❑ <attribute name>: the name of an attribute of *R*
  - ❑ <comparison op>: normally one of the operators $\{=, <, \leq, >, \geq, \neq\}$
    - Ordered values $\{=, <, \leq, >, \geq, \neq\}$ (numeric and date) vs unordered values $\{=, \neq\}$ (e.g., Color={'red', 'blue', …}
  - ❑ <constant value>: a constant value from the attribute domain

- Clauses can be connected by **AND** and **OR**

$$\sigma_{(\text{Dno}=4 \textbf{ AND } \text{Salary}>25000) \textbf{ OR } (\text{Dno}=5 \textbf{ AND } \text{Salary}>30000)}(\text{EMPLOYEE})$$

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5

Properties of SELECT Operation

❑ The SELECT Operation is *Commutative*

- A sequence of SELECTs can be applied in an order

$$\sigma_{<cond1>}(\sigma_{<cond2>}(\ldots(\sigma_{<condn>}(R))\ldots)) = \sigma_{<cond1>\ \textbf{AND}\ <cond2>\ \textbf{AND}\ldots\textbf{AND}\ <condn>}(R)$$

- A sequence of SELECTs can be combined into a single SELECT

$$\sigma_{<cond1>}(\sigma_{<cond2>}(R)) = \sigma_{<cond2>}(\sigma_{<cond1>}(R))$$

❑ Correspondence to SQL

- SELECT operation is different from the SELECT clause of SQL

$\sigma_{\text{Dno=4 AND Salary>25000}}$ (EMPLOYEE) ⟺

```
SELECT  *
FROM    EMPLOYEE
WHERE   Dno=4 AND Salary>25000;
```

Containing all attributes of *R*

The SELECT condition is specified in the WHERE clause of a query

Unary Relational Operations: PROJECT

What is the PROJECT Operation?

- To select certain columns from the table and discards the other columns
  - i.e., a **filter** that keeps only the columns, while discards the others

The PROJECT operator (pi) $\pi_{\text{<attribute list>}}(R)$ A relational algebra expression (relation)

The desired list of attributes to retrieve

- Result of the PROJECT Operation
  - The relation has only the attributes specified in <attribute list>
    - i.e., its degree is the same as the number of attributes in <attribute list>

$\pi_{\text{Lname, Fname, Salary}}(\text{EMPLOYEE})$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Vertical filtering

Properties of PROJECT Operation

❑ Duplicate Elimination

- Duplicate tuples are likely to occur *if the attribute list includes only nonkey attributes*
- The PROJECT operation removes any duplicate tuples
  - ❑ i.e., the result of the PROJECT operation is a "***set***" of distinct tuples

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

$\pi_{Sex,\ Salary}(EMPLOYEE)$

Sex	Salary
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

Set of distinct tuples

❑ Correspondence to SQL

$\pi_{Sex,\ Salary}(EMPLOYEE)$ ⟺

```
SELECT   DISTINCT Sex, Salary
FROM     EMPLOYEE
```

The PROJECT attribute list is specified in the SELECT clause of a query

RENAME Operation

What is the RENAME Operation?

- To rename either the relation name or the attribute names, or both

$$\rho_{S(B1,\,B2,\,\ldots\,,\,Bn)}(R) \quad \text{or} \quad \rho_{S}(R) \quad \text{or} \quad \rho_{(B1,\,B2,\,\ldots\,,\,Bn)}(R)$$

Both the relation and its attributes | The relation only | The attributes only

- The symbol ρ (rho): the RENAME operator
- S: the new relation name
- B_i: the new attribute name ($1 \leq i \leq n$, n is the degree of R)

$$\text{TEMP} \leftarrow \sigma_{\text{Dno}=5}(\text{EMPLOYEE})$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
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Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

TEMP

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston,TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston,TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

R

First_name	Last_name	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

$$\text{R} \leftarrow \rho_{(\text{First_nam,Last_name,Salary})}(\pi_{\text{Fname,Lname,Salary}}(\text{TEMP}))$$

Correspondence to SQL

❑ Relational Algebra Expression vs SQL Query

- A single query typically represents a complex relational algebra expression

Q. Retrieve the first name, last name, and salary of all employees who work in department number 5

```
SELECT   E.Fname AS First_name, E.Lname AS Last_name, E.Salary AS Salary
FROM     EMPLOYEE AS E
WHERE    E.Dno=5;
```

Renaming in SQL is accomplished by aliasing using **AS**



$$TEMP \leftarrow \sigma_{Dno=5}(EMPLOYEE)$$

$$R(First_name, Last_name, Salary) \leftarrow \pi_{Fname, Lname, Salary}(TEMP)$$

Sequences of Relational Operations

❑ What If Need to Apply Several Relational Algebra Operations?

Q. Retrieve the first name, last name, and salary of all employees who work in department number 5

Method 1:

$$DEP5_EMPS \leftarrow \sigma_{Dno=5}(EMPLOYEE)$$
$$RESULT \leftarrow \pi_{Fname,\ Lname,\ Salary}(DEP5_EMPS)$$

Explicitly show the sequence of operations by **giving a name to each intermediate relation**

Method 2:

$$\pi_{Fname,\ Lname,\ Salary}(\sigma_{Dno=5}(EMPLOYEE))$$

Write the operations as a **relational algebra expression** by nesting the operations

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEP5_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston,TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston,TX	M	40000	888665555	5
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble,TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

RESULT

Fname	Lname	Salary
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

It is usually simpler to break down a complex sequence of operations by specifying intermediate result relations than to write a single relational algebra expression

Outline

- Introduction to Relational Algebra
- Unary Relational Operations
- **Set Operations**



- **Binary Relational Operations**
- **Additional Relational Operations**

Set Operations

❑ Set Operations as Relational Algebra Operations

Q. Retrieve the SSNs of all employees who either work in department 5 or directly supervise an employee who works in department 5

$$\text{DEP5_EMPS} \leftarrow \sigma_{\text{Dno}=5}(\text{EMPLOYEE})$$
$$\text{RESULT1} \leftarrow \pi_{\text{Ssn}}(\text{DEP5_EMPS})$$
$$\text{RESULT2(Ssn)} \leftarrow \pi_{\text{Super_ssn}}(\text{DEP5_EMPS})$$
$$\text{RESULT} \leftarrow \text{RESULT1} \cup \text{RESULT2}$$

RESULT1 and RESULT2 are *type-compatible*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

RESULT1

Ssn
123456789
333445555
666884444
453453453

RESULT2

Ssn
333445555
888665555

RESULT

Ssn
123456789
333445555
666884444
453453453
888665555

The SSN value '33344555' appears only once

❑ Types of Set Operations

- UNION, INTERSECTION, SET DIFFERENCE (MINUS or EXCEPT)
  - ❑ Those operations can be applied relations that are type-compatible
    - Two relations $R(A_1, \ldots, A_n)$ and $S(B_1, \ldots, B_n)$ are said to be *type compatible*:

If $\text{dom}(A_i) = \text{dom}(B_i)$ for $1 \leq i \leq n$

The two relations have the same number of attributes, and each corresponding pair of attributes has the same domain

Set Operations: UNION and INTERSECTION

❑ UNION Operation

- The result is a relation including all tuples that are either **in *R*** or **in *S*** or **in both**
  - ❑ Denoted by $R \cup S$, *duplicate tuples are eliminated*

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT and INSTRUCTOR are type-compatible

❑ INTERSECTION Operation

- The result is a relation including all tuples that are in both *R* and *S*
  - ❑ Denoted by $R \cap S$

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

STUDENT ∪ INSTRUCTOR

Fn	Ln
Susan	Yao
Ramesh	Shah

STUDENT ∩ INSTRUCTOR

❑ Properties of UNION and INTERSECTION

- $R \cup S = S \cup R, R \cap S = S \cap R$ (i.e., *Commutative*)
- $R \cup (S \cup T) = (S \cup R) \cup T$, $R \cap (S \cap T) = (S \cap R) \cap T$ (i.e., *Associative*)

Set Operations: SET DIFFERENCE

SET DIFFERENCE (MINUS) Operation

- The result is a relation including all tuples that are in *R* but not in *S*
  - Denoted by $R - S$

STUDENT

Fn	Ln
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR

Fname	Lname
John	Smith
Ricardo	Browne
Susan	Yao
Francis	Johnson
Ramesh	Shah

STUDENT – INSTRUCTOR

Fn	Ln
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

INSTRUCTOR – STUDENT

Fname	Lname
John	Smith
Ricardo	Browne
Francis	Johnson

The MINUS Operation is Not *Commutative*

- $R - S \neq S - R$

CARTESIAN PRODUCT

What is the CARTESIAN PRODUCT Operation?

- A binary set operation, which can be applied to not type-compatible relations
  - a.k.a **CROSS PRODUCT** or **CROSS JOIN**
- Produce new tuples by concatenating *all possible combinations of tuples*
  - Denoted by $R(A_1, \dots, A_n) \times S(B_1, \dots, B_m) \rightarrow Q(A_1, \dots, A_n, B_1, \dots, B_m)$
    - The number of attributes: $n + m$
    - The number of tuples: $|R| * |S|$ (all possible combinations)

Usage of the CARTESIAN PRODUCT

- This operation alone is generally *meaningless*
- Useful only when followed by a *selection matching certain values of attributes*

CARTESIAN PRODUCT: Example

❑ When the CARTESIAN PRODUCT Useful?

Q. Retrieve a list of names of each female employee's dependents

$$\text{FEMALE_EMPS} \leftarrow \sigma_{\text{Sex='F'}}(\text{EMPLOYEE})$$
$$\text{EMPNAMES} \leftarrow \pi_{\text{Fname, Lname, Ssn}}(\text{FEMALE_EMPS})$$
$$\text{EMP_DEPENDENTS} \leftarrow \text{EMPNAMES} \times \text{DEPENDENT}$$
$$\text{ACTUAL_DEPENDENTS} \leftarrow \sigma_{\text{Ssn=Essn}}(\text{EMP_DEPENDENTS})$$
$$\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Dependent_name}}(\text{ACTUAL_DEPENDENTS})$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

FEMALE_EMPS

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	1968-07-19	3321Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291Berry, Bellaire, TX	F	43000	888665555	4
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

3 tuples

X

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

7 tuples

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	…
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	…
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	…
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	…
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	…
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	…
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	…
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	…
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	…
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	…
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	…
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	…
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	…
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	…
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	…
Joyce	English	453453453	333445555	Alice	F	1986-04-05	…
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	…
Joyce	English	453453453	333445555	Joy	F	1958-05-03	…
Joyce	English	453453453	987654321	Abner	M	1942-02-28	…
Joyce	English	453453453	123456789	Michael	M	1988-01-04	…
Joyce	English	453453453	123456789	Alice	F	1988-12-30	…
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	…

3*7 tuples

Every tuple from EMPNAMES is combined with every tuple from DEPENDENT

This result is not that meaningful!

CARTESIAN PRODUCT: Example (Cont.)

❑ When the CARTESIAN PRODUCT Useful?

Q. Retrieve a list of names of each female employee's dependents

$$\text{FEMALE_EMPS} \leftarrow \sigma_{\text{Sex='F'}}(\text{EMPLOYEE})$$
$$\text{EMPNAMES} \leftarrow \pi_{\text{Fname, Lname, Ssn}}(\text{FEMALE_EMPS})$$
$$\text{EMP_DEPENDENTS} \leftarrow \text{EMPNAMES} \times \text{DEPENDENT}$$
$$\text{ACTUAL_DEPENDENTS} \leftarrow \sigma_{\text{Ssn=Essn}}(\text{EMP_DEPENDENTS})$$
$$\text{RESULT} \leftarrow \pi_{\text{Fname, Lname, Dependent_name}}(\text{ACTUAL_DEPENDENTS})$$

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

We want to combine a female employee tuple only with her particular dependents

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

RESULT

Fname	Lname	Dependent_name
Jennifer	Wallace	Abner

CARTESIAN PRODUCT followed by SELECT combines related tuples from two relations (→ **JOIN**)

Outline

- Introduction to Relational Algebra
- Unary Relational Operations
- Set Operations
- **Binary Relational Operations**
- **Additional Relational Operations**



Binary Relational Operations: JOIN

❑ What is the JOIN Operation?

- Combine *related tuples* from two relations into single "longer" tuples

The JOIN operator (bowtie) $R \bowtie_{\text{<join condition>}} S$

- Result of the JOIN Operation: $Q(A_1, \ldots, A_n, B_1, \ldots, B_m)$
  - Contain only *the combinations of tuples satisfying the join condition*
    - The join condition is evaluated for each tuple combination; then included it if it satisfies the condition

Q. Retrieve the name of the manager of each department:

$$\text{DEPT_MGR} \leftarrow \text{DEPARTMENT} \bowtie_{\text{Mgr_ssn=Ssn}} \text{EMPLOYEE}$$
$$\text{RESULT} \leftarrow \pi_{\text{Dname, Lname, Fname}}(\text{DEPT_MGR})$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

Mgr_ssn is a **foreign key** of the DEPARTMENT that references Ssn, the **primary key** of the EMPLOYEE

DEPT_MGR

Dname	Dnumber	Mgr_ssn	···	Fname	Minit	Lname	Ssn	···
Research	5	333445555	···	Franklin	T	Wong	333445555	···
Administration	4	987654321	···	Jennifer	S	Wallace	987654321	···
Headquarters	1	888665555	···	James	E	Borg	888665555	···

Combine each department tuple with the employee tuple

CARRESIAN PRODUCT vs JOIN Operation

❑ Comparison of JOIN with CARTESIAN PRODUCT

- The JOIN can be specified as a CARTESIAN PRODUCT followed by SELECT

$$\text{EMP_DEPENDENTS} \leftarrow \text{EMPNAMES} \times \text{DEPENDENT}$$
$$\text{ACTUAL_DEPENDENTS} \leftarrow \sigma_{\text{Ssn=Essn}}(\text{EMP_DEPENDENTS}) \quad \Longleftrightarrow \quad \text{ACTUAL_DEPENDENTS} \leftarrow \text{EMPNAMES} \bowtie_{\text{Ssn=Essn}} \text{DEPENDENT}$$

These two operations can be replaced with **a single JOIN operation**

EMPNAMES

Fname	Lname	Ssn
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

EMP_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	...
Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	...
Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	...
Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	...
Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	...
Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	...
Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	...
Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	...
Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	...
Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...
Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	...
Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	...
Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	...
Joyce	English	453453453	333445555	Alice	F	1986-04-05	...
Joyce	English	453453453	333445555	Theodore	M	1983-10-25	...
Joyce	English	453453453	333445555	Joy	F	1958-05-03	...
Joyce	English	453453453	987654321	Abner	M	1942-02-28	...
Joyce	English	453453453	123456789	Michael	M	1988-01-04	...
Joyce	English	453453453	123456789	Alice	F	1988-12-30	...
Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	...

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

Generalized JOIN Operation

❑ Join Operation with General Join Conditions

- Made up of multiple Boolean expressions, combined with **AND**

$R \bowtie_{\text{<join condition>}} S$

<condition> **AND** <condition> **AND ... AND** <condition>

Include tuples only if all conditions are satisfied

- Each <condition> is of the form "$A_i\ \theta\ B_j$"
  - A_i: an attribute of *R*
  - B_j: an attribute of *S*
  - θ (**theta**): one of the comparison operations $\{=, <, \leq, >, \geq, \neq\}$

A JOIN operation with such general join conditions is called a **THETA JOIN**

↓

Two widely used variations of JOIN:
EQUIJOIN and **NATURAL JOIN**

Variations of JOIN: EQUIJOIN

What is the EQUIJOIN?

- A JOIN operation, where the only comparison operator used is **"="**
- One of more pairs of attributes should have identical values in every tuple
  - The join condition specified on these attributes *requires the values to be identical*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

DEPT_MGR ← DEPARTMENT ⋈ Mgr_ssn=Ssn EMPLOYEE

DEPT_MGR

Dname	Dnumber	Mgr_ssn	...	Fname	Minit	Lname	Ssn	...
Research	5	333445555	...	Franklin	T	Wong	333445555	...
Administration	4	987654321	...	Jennifer	S	Wallace	987654321	...
Headquarters	1	888665555	...	James	E	Borg	888665555	...

One of each pair of attributes with identical values is *unnecessary*
→ NATURAL JOIN

ACTUAL_DEPENDENTS ← EMPNAMES ⋈ Ssn=Essn DEPENDENT

ACTUAL_DEPENDENTS

Fname	Lname	Ssn	Essn	Dependent_name	Sex	Bdate	...
Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	...

Variations of JOIN: NATURAL JOIN

❑ What is the NATURAL JOIN?

- A JOIN operation *to get rid of the redundant attribute* in an EQUIJOIN condition
  - ❑ *The join attributes should have the same name* in both relations; otherwise, a *renaming is needed*
  - ❑ Denoted by $*$

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

Two relations have the same names

DEPT_LOCS ← DEPARTMENT * DEPT_LOCATIONS

DEPT_LOCS

Dname	Dnumber	Mgr_ssn	Mgr_start_date	Location
Headquarters	1	888665555	1981-06-19	Houston
Administration	4	987654321	1995-01-01	Stafford
Research	5	333445555	1988-05-22	Bellaire
Research	5	333445555	1988-05-22	Sugarland
Research	5	333445555	1988-05-22	Houston

Only one join attribute

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

Rename first

DEPT ← $\rho_{(Dname, Dnum, Mgr_ssn, Mgr_start_date)}$(DEPARTMENT)

PROJ_DEPT ← PROJECT * DEPT

PROJ_DEPT

Pname	Pnumber	Plocation	Dnum	Dname	Mgr_ssn	Mgr_start_date
ProductX	1	Bellaire	5	Research	333445555	1988-05-22
ProductY	2	Sugarland	5	Research	333445555	1988-05-22
ProductZ	3	Houston	5	Research	333445555	1988-05-22
Computerization	10	Stafford	4	Administration	987654321	1995-01-01
Reorganization	20	Houston	1	Headquarters	888665555	1981-06-19
Newbenefits	30	Stafford	4	Administration	987654321	1995-01-01

Only one join attribute

Properties of JOIN Operation

❑ The Size of the Result of JOIN Operation

$$Q \leftarrow R \bowtie_{\text{<join condition>}} S \qquad 0 \leq |Q| \leq |R| * |S|$$

- (Minimum) If no combination of tuples satisfies the join condition
  - ❑ → The result is an empty relation
- (Maximum) If all combinations of tuples satisfy the condition or no join condition
  - ❑ → The result includes all possible combinations of tuples (i.e., **CARTESIAN PRODUCT**)

❑ Join Selectivity of a Join Condition

$$\frac{\text{Expected Size of the Join Result}}{|R| * |S|}$$

It can be used to estimate the cost of query plans in query optimization

Binary Relational Operations: DIVISION

What is the DIVISION Operation?

- Useful for a special kind of query that sometimes occurs in DB applications
  - This operation, denoted by ÷, can be performed with the basic relational algebra operations
- Example:
  - Retrieve the names of employees who work on ***all the projects*** that 'John Smith' works on

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

All the projects,
not a single specific project

Example of DIVISION Operation

❑ How to Specify the DIVISION Operation in Relational Algebra?

Q.Retrieve the names of employees who work on ***all*** the projects that 'John Smith' works on

Find all the projects that 'John Smith' works on

$$SMITH \leftarrow \sigma_{Fname='John' \text{ AND } Lname='Smith'}(EMPLOYEE)$$
$$SMITH_PNOS \leftarrow \pi_{Pno}(WORKS_ON \bowtie_{Essn=Ssn} SMITH)$$

Create a relation about employees working on projects

$$SSN_PNOS \leftarrow \pi_{Essn,\ Pno}(WORKS_ON)$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

SMITH_PNOS

Pno
1
2

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

SSN_PNOS

Essn	Pno
123456789	1
123456789	2
666884444	3
453453453	1
453453453	2
333445555	2
333445555	3
333445555	10
333445555	20
999887777	30
999887777	10
987987987	10
987987987	30
987654321	30
987654321	20
888665555	20

<Pno, Essn> Relation

Example of DIVISION Operation (Cont.)

❑ How to Specify the DIVISION Operation in Relational Algebra?

Q.Retrieve the names of employees who work on ***all*** the projects that 'John Smith' works on

$$SSNS(Ssn) \leftarrow SSN_PNOS \div SMITH_PNOS$$

$$RESULT \leftarrow \pi_{Fname,\ Lname}(SSNS * EMPLOYEE)$$

SSN_PNOS

Essn	Pno
123456789	1
123456789	2
666884444	3
453453453	1
453453453	2
333445555	2
333445555	3
333445555	10
333445555	20
999887777	30
999887777	10
987987987	10
987987987	30
987654321	30
987654321	20
888665555	20

÷

SMITH_PNOS

Pno
1
2

=

SSNS

Ssn
123456789
453453453

*

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

By applying the DIVISION operation, the desired SSNs are retrieved

Form of the DIVISION Operation

How to Use DIVISION Operation?

- Formally, the DIVISION operation is applied to two relations *R(Z)* and *S(X)*
  - $R(Z) \div S(X), X \subseteq Z$ (the attributes of S are a subset of the attributes of R)
- The result of this operation is a relation *T(Y)*
  - $Y = Z - X$ (i. e. , $X \cup Y = Z$)

$Z = \{A, B\}$

R

A	B
a1	b1
a2	b1
a3	b1
a4	b1
a1	b2
a3	b2
a2	b3
a3	b3
a4	b3
a1	b4
a2	b4
a3	b4

Numerator

$\div$

$X = \{A\}$

S

A
a1
a2
a3

Denominator

$=$

$Y = \{B\}$

T

B
b1
b4

$T \leftarrow R \div S$

The tuples in the denominator relation S restrict the numerator relation R by selecting those tuples in the result matching all values present in the denominator.

Complete Set of Relational Algebra Operations

❑ A Complete Set of Relational Algebra Operations

- The following six operations are elements of this set

$$\{\sigma, \pi, \cup, \rho, -, \times\}$$

  - ❑ Any other original relational algebra operations can be expressed as a sequence of these operations

INTERSECTION

$$R \cap S \equiv (R \cup S) - ((R - S) \cup (S - R))$$

JOIN

$$R \bowtie_{<\text{condition}>} S \equiv \sigma_{<\text{condition}>}(R \times S)$$

JOIN = **CARTESIAN PRODUCT** followed by **SELECT**

Strictly, **INTERSECTION** and **JOIN** are not required.

However, they are important to include as separate operations since they are very commonly used in DB applications

NATURAL JOIN can be specified as a **CARTESIAN PRODUCT**, preceded by **RENAME** and followed by **SELECT** and **PROJECT**

Outline

- ❑ Introduction to Relational Algebra
- ❑ Unary Relational Operations
- ❑ Set Operations
- ❑ Binary Relational Operations
- ❑ **Additional Relational Operations**



Additional Relational Operations

The Basic Operations Are Not Enough

- Some common DB Requests cannot be performed with the basic operations

Examples

- Requests involving *aggregate functions* on collections of values from the database
  - To summarize information from the database tuples such as **SUM**, **AVERAGE**, **MAXIMUM**, and **COUNT**
- Requests involving *grouping* the tuples in a relation by the value of some attributes
  - Then, apply an aggregate function *independently to each group*
- Requests involving *outer join operations* on two relations

Additional relational operations are needed to express these requests
→ Enhancing the expressive power of the original relational algebra!

Aggregate Functions and Grouping

How to Specify Aggregate Functions in Relational Algebra?

- Mathematical aggregate functions on collections of values from database
  - e.g., AVERAGE, SUM, MAXIMUM, MINIMUM, and COUNT

Q1. Retrieve the average and total salary of all employees

Q2. Retrieve the total number of employees

How to Specify Requests Involving Grouping the Tuples?

- Grouping tuples by value of some attributes and applying an aggregate function

Q3. Group EMPLOYEE tuples by Dno,
so that each group includes the tuples for employees working in the same department.

Q3-1. Retrieve the average and total salary of employees in each department
Q3-2. Retrieve the number of employees in each department

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Form of Aggregate Functions and Grouping

❑ How to Formally Define an Aggregate Function?

$$\text{<grouping attributes>}\ \Im\ \text{<function list>}\ (R)$$

- Symbol $\Im$ (Script F): the aggregate function operator
- Grouping attributes: a list of attributes of the relation *R* to be used for grouping
- Function list: a list of (<function> <attribute>) pairs
  - <function>: one of the functions such as SUM, AVERAGE, MAXIMUM, MINIMUM, and COUT
  - <attribute>: an attribute of the relation specified by *R*

❑ Output of an Aggregate Function

- A relation involving the grouping attributes with one attribute for each element in the function list

Aggregate Functions and Grouping: Examples

❑ Examples of AGGREGATE FUNCTION Operations

Q. Retrieve each department number, the number of employees in the department, and their average salary, while renaming the resulting attributes as indicated below

$$\rho_{R(\text{Dno, No_of_employees, Average_sal})}(\text{Dno } \mathfrak{I} \text{ COUNT Ssn, AVERAGE Salary } (\text{EMPLOYEE})).$$

Renaming the attributes Grouping attribute Function list

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Grouping attribute + one element for each function

R

Dno	No_of_employees	Average_sal
5	4	33250
4	3	31000
1	1	55000

Q. What if no renaming operation is specified?

Q. What if no grouping attribute is specified?

Aggregate Functions and Grouping: Examples

❑ Examples of AGGREGATE FUNCTION Operations

Q. Retrieve ~~each department number~~, the number of employees ~~in the department~~, and their average salary, ~~while renaming the resulting attributes as indicated below~~

$$_{Dno}\mathfrak{I}_{\text{COUNT Ssn, AVERAGE Salary}}(\text{EMPLOYEE}).$$

$$\mathfrak{I}_{\text{COUNT Ssn, AVERAGE Salary}}(\text{EMPLOYEE}).$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

The concatenation of the function name with the attribute name:
Form of <function>_<attribute>

Dno	Count_ssn	Average_salary
5	4	33250
4	3	31000
1	1	55000

Count_ssn	Average_salary
8	35125

The functions are applied to all the tuples in the relation
Thus, the resulting relation has a ***single tuple only***

Aggregate Functions and Grouping: Properties

Duplicate Tuples and NULL in Aggregate Functions

- Duplicates are *not eliminated* when an aggregate function is applied
  - e.g., compute the sum of salaries or the average of salaries in each department
- NULL values are *not considered* in the aggregation

Relational Algebra As a Closed Mathematical System

- The result of an aggregate function is also a "relation", not a scalar number
  - Even if it has a single value or tuple

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

Count_ssn	Average_salary
8	35125

This is a relation not a scalar number

OUTER JOIN Operations

❑ Results of the JOIN Operations (EQUIJOIN & NATURAL JOIN)

- Only tuples satisfying the join conditions are included in the resulting relation
  - Tuples with *no matching tuple are eliminated* (i.e., **inner join**)

What if some users want the result of the JOIN to include all the tuples without the matching tuples?

Example: What if we want a list of all employee names as well as the name of the departments they manage if they happen to manage a department; if they do not manage one

❑ How to Specify OUTER JOIN Operations in Relational Algebra?

- **OUTER JOIN** operations can be used for those cases
  - Users want to keep all the tuples in **one** or **more** of the relations in the result of the JOIN
  - Regardless of whether they have matching tuples in the other relation

OUTER JOIN Operations: Example

❑ How to Specify OUTER JOIN Operations in Relational Algebra?

Q. Retrieve all employee names as well as the name of the departments they manage if they happen to manage a department; if they do not manage one

$$TEMP \leftarrow (EMPLOYEE \overset{\bowtie}{} {}_{Ssn=Mgr_ssn} DEPARTMENT)$$
$$RESULT \leftarrow \pi_{Fname,\ Minit,\ Lname,\ Dname}(TEMP)$$

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
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Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

RESULT

Fname	Minit	Lname	Dname
John	B	Smith	NULL
Franklin	T	Wong	Research
Alicia	J	Zelaya	NULL
Jennifer	S	Wallace	Administration
Ramesh	K	Narayan	NULL
Joyce	A	English	NULL
Ahmad	V	Jabbar	NULL
James	E	Borg	Headquarters

Padded with NULL values

Including every tuple in the first (**left**) relation

❑ Types of OUTER JOIN Operations



LEFT OUTER JOIN



RIGHT OUTER JOIN



FULL OUTER JOIN

Takeaways: Relational Algebra Operations

	OPERATION	PURPOSE	NOTATION
Unary Relational Operations	SELECT	Selects all tuples that satisfy the selection condition from a relation R.	$\sigma_{\text{<selection condition>}}(R)$
	PROJECT	Produces a new relation with only some of the attributes of R, and removes duplicate tuples.	$\pi_{\text{<attribute list>}}(R)$
Binary Relational Operations	THETA JOIN	Produces all combinations of tuples from R_1 and R_2 that satisfy the join condition.	$R_1 \bowtie_{\text{<join condition>}} R_2$
	EQUIJOIN	Produces all the combinations of tuples from R_1 and R_2 that satisfy a join condition with only equality comparisons.	$R_1 \bowtie_{\text{<join condition>}} R_2$, OR $R_1 \bowtie_{\text{(<join attributes 1>), (<join attributes 2>)}} R_2$
	NATURAL JOIN	Same as EQUIJOIN except that the join attributes of R_2 are not included in the resulting relation; if the join attributes have the same names, they do not have to be specified at all.	$R_1 *_{\text{<join condition>}} R_2$, OR $R_1 *_{\text{(<join attributes 1>), (<join attributes 2>)}} R_2$ OR $R_1 * R_2$
Set Operations	UNION	Produces a relation that includes all the tuples in R_1 or R_2 or both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cup R_2$
	INTERSECTION	Produces a relation that includes all the tuples in both R_1 and R_2; R_1 and R_2 must be union compatible.	$R_1 \cap R_2$
	DIFFERENCE	Produces a relation that includes all the tuples in R_1 that are not in R_2; R_1 and R_2 must be union compatible.	$R_1 - R_2$
	CARTESIAN PRODUCT	Produces a relation that has the attributes of R_1 and R_2 and includes as tuples all possible combinations of tuples from R_1 and R_2.	$R_1 \times R_2$

Takeaways: Additional Relational Operations

❑ DIVISION Operation

- $R(Z) \div S(X) = T(Y), X \subseteq Z, Y = Z - X$
  - ❑ The attributes of S are a subset of the attributes of R

❑ Aggregate and Grouping Functions

$$<\text{grouping attributes}> \Im <\text{function list}> (R)$$

- ❑ Output: A relation with the grouping attributes with one attribute for each element in the function list

❑ Outer Join Operations



LEFT OUTER JOIN



RIGHT OUTER JOIN



FULL OUTER JOIN

Q&A:
Relational Algebra